

Verificación heterogénea y acceso a la IA

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Abstract

Replace. One paragraph, at most 150 words: the question, the model in one sentence, the main result *with its condition*, and what the Lean formalization does and does not certify.

1 Introduction

Replace. The question, why it matters, and your answer. State the main result in words in the first page. Say which paper you extend and which assumption you relax (Track A), or which thesis model you formulate (Track B). End with what is new relative to the closest paper. The toy model below only shows the mechanics of this template; none of it is yours to keep.

2 Related literature

Replace. Two or three paragraphs. Position the paper against the closest work, cite the exact result you depart from (proposition and page), and check the editorial status of every preprint you cite.

El proyecto extiende la Proposición 6 de [Ide and Talamàs \(2025\)](#), p. 27 de arXiv v11. El modelo de esfuerzo que sigue es únicamente el ejemplo pendiente de la plantilla; no es un resultado de este proyecto.

3 Model

Replace. Primitives, timing, information, and the agent’s problem written formally: what is maximised, over which variable, subject to which constraints. Number every assumption; a result may only use numbered assumptions.

A worker chooses effort $e \geq 0$. Output is ae , where a is productivity with AI assistance, and effort costs $\frac{c}{2}e^2$.

Assumption 1. $a > 0$ and $c > 0$.

The worker solves

$$\max_{e \geq 0} ae - \frac{c}{2}e^2. \tag{1}$$

*Universidad del Pacífico. Repository: <https://github.com/sbvasquezg-ux/ai-project>. Track A: extensión de Ide y Talamàs (2025). Plantilla pendiente, entrega 26 de noviembre de 2026.

4 Results

Replace. Every proposition with *all* its conditions and a complete proof. A proof may be moved to Appendix A, but the statement and the idea of the proof stay here.

Proposition 1. *Under Assumption 1, problem (1) has the unique solution $e^* = a/c$, and $\partial e^*/\partial a = 1/c > 0$.*

Proof. The objective is twice differentiable with second derivative $-c < 0$, so it is strictly concave and has at most one maximiser. The first-order condition $a - ce = 0$ gives $e = a/c$, which is positive under Assumption 1, so the constraint $e \geq 0$ does not bind and $e^* = a/c$. Differentiating, $\partial e^*/\partial a = 1/c > 0$. $\square$

5 Simulations

Replace. A symbolic or numerical check of every result (SymPy, Monte Carlo, or both), produced by the scripts in `code/`. Say what the simulation can and cannot show: it checks a derivation on a grid, it does not prove it.

Replace. Figura y simulación del paper final pendientes. El ejemplo de esfuerzo de la plantilla fue retirado de `code/`. Los chequeos actuales del topic están en `code/verify.py`.

6 Discussion and limitations

Replace. What the result depends on, which assumption does the work, and what you could not prove. A stated limitation is worth more than a hidden one.

7 Conclusion

Replace. One paragraph. No new results.

References

Enrique Ide and Eduard Talamàs. Artificial intelligence in the knowledge economy. *Journal of Political Economy*, 133(12):3762–3800, 2025. doi: 10.1086/737233. URL <https://arxiv.org/abs/2312.05481v11>. Referencia de paginación: arXiv:2312.05481v11, Proposición 6, p. 27. Versión v12: p. 28.

A Omitted proofs

Replace. Proofs and derivations moved out of the body. The handwritten, step-by-step version of every derivation lives in `hand/`.

B Lean formalization

Replace. One row per numbered result of the paper. “Status” is one of: *proved* (no **sorry**, no added hypothesis), *proved under an added hypothesis* (state it), or *open* (state the exact blocker). Report the paper commit that was formalized and the output of `paper_contribution.py check <Folder> -fast`.

Paper	Lean declaration	File in <code>lean/</code>	Status
Proposition 1	<code><Folder>.prop_effort</code>	<code>MainTheorems.lean</code>	<i>example row</i>

C AI collaboration log

Replace. For each relevant contribution of a language model: what it claimed, your verdict (correct, incorrect, or not verifiable) with the reason, and how you checked it. The raw prompts and answers are in `prompts.md`.

Claim by the model	Verdict	How it was checked
<i>example:</i> “ $e^* = a/c$ for any a ”	Incorrect as stated	Fails for $a \leq 0$: the corner $e^* = 0$ binds. Hand derivation, p. 1.